

Dominic Pearce

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Software Engineer & Bioinformatician.

Experience

Principal Bioinformatics Developer

Fios Genomics

📅 Nov 2022 — February 2026 📍 Edinburgh, UK

- Providing technical leadership and directing the R&D strategy.
- Management of the R&D team.
- Design, development and orchestration of 25+ internal R and python packages, in line with best practices.

Senior Bioinformatics Developer

Fios Genomics

📅 May 2021 — Oct 2022 📍 Edinburgh, UK

- Development of novel internal software in line with best practices.
- Maintenance of 20+ standalone R and python packages.
- Providing technical support to the operations team.

Bioinformatician

Fios Genomics

📅 Nov 2018 — Apr 2020 📍 Edinburgh, UK

- Performing bioinformatic and statistical analysis of various -omics data types.
- Communicating with relevant stakeholders to report and interpret results and define future studies.

References available on request

Education

University of Edinburgh

📅 Sep 2014 — Jun 2018 📍 Edinburgh, UK

PhD at Institute of Genetics and Cancer. Supervisor: Dr. Andrew Sims. Thesis title: Breast cancer heterogeneity within the neoadjuvant window.

University of Leicester

📅 Sep 2011 — Jun 2012 📍 Leicester, UK

MSc. (distinction) in Cancer Cell and Molecular Biology.

University of Nottingham

📅 Sep 2007 — Jun 2010 📍 Nottingham, UK

BSc. 2.1 in Biochemistry.

Selected publications

- Pearce, D. A. et al. Tumour sampling method can significantly influence gene expression profiles derived from neoadjuvant window studies. *Sci. Rep.*
- Pearce, D. A. et al. Continuous Biomarker Assessment by Exhaustive Survival Analysis. *bioRxiv*
- Moir, N, Pearce, D. A, Langdon, S. P & Simpson, I. The significance of molecular heterogeneity in breast cancer batch correction and dataset integration. *Breast Cancer Research.*
- Ross, K. E., Pearce, D. A, McMillan, K. M. What Patterns in Online Classified Puppy Advertisements Can Tell Us about the Current UK Puppy Trade. *Animals.*

Technical Expertise

R	●●●●●
Python	●●●●●
Bioinformatics	●●●●●
Statistics	●●●●●
ML & AI	●●●●●

Skills/Exposure

- Team leadership • Omics sequencing technologies • Single-cell analysis • Antisense oligonucleotide (ASO) design • Machine learning, AI and LLMs • Bayesian methods

Methodology/Approach

- Agile • Test-driven development • Literate programming • Reproducible environments

Tools

- Bioconductor • Git • GitLab & GitHub • Unix • Quarto • Shiny • Vim • tidyverse • Docker

Selected projects

Internal semantic document search

To support cross-team collaboration I designed and built a python package for transformer-based natural language processing (NLP) of internal documents. This was hosted as a Dockerised Shiny app, allowing teams to semantically search legacy proposals, reports and similar documents, making this information easily available for all employees.

Interests

- Over-the-board chess
- 5-a-side football
- My dogs, cat and children

Antisense oligonucleotide design pipeline

In collaboration with a large CRO (Charles River Laboratories) I designed and built an antisense oligonucleotide (ASO) design pipeline, allowing all possible candidate ASOs to be exhaustively assessed **in silico**, in terms of their target binding capacity, non-specific binding potential, etc. This allowed for unsuitable candidate ASOs to be failed early and experimental validation performed more cost-efficiently.